

Anatomy of the Mediastinum & Its Disorders

The Big Picture

The mediastinum = the central chest compartment between the two pleural sacs, **sternum** → **vertebrae**, **thoracic inlet** → **diaphragm**. It packs the heart, great vessels, trachea, oesophagus, thymus, nerves, thoracic duct and nodes. Exam gold: **which compartment, what mass lives there**, plus mediastinitis, shift and widening.

Definition & Boundaries

- Partition between right/left pleural sacs; contains all the intermediate-compartment thoracic structures.
- Bounded: pleura (lateral), **sternum (anterior)**, **vertebrae (posterior)**, thoracic inlet (superior), diaphragm (inferior).

Classifications

- **Felson (3 compartments)**: Anterior (sternum→back of heart/great vessels), Middle (to 1 cm behind anterior vertebral border), Posterior (behind that).
- **ITMIG (CT-based, modern)**: **Prevascular (anterior)**, **Visceral (middle)**, **Paravertebral (posterior)**.
- **Superior/inferior** division: transverse plane from **sternal angle** → **T4-5 disc**.

Compartments & Their Masses

- **Prevascular / Anterior** — sternum to great vessels/pericardium. Contents: thymus, fat, nodes, internal mammary. **~50% of all mediastinal masses**. Masses = the **4 T's**: **Thymoma**, **"Terrible" lymphoma**, **Thyroid (substernal)**, **Teratoma/germ cell**.
- **Visceral / Middle** — heart, pericardium, aorta + branches, SVC, trachea, oesophagus, pulmonary arteries, thoracic duct. Masses: **lymphoma**, **metastatic lymphadenopathy**, aortic aneurysm, **bronchogenic/foregut cysts**, oesophageal/tracheal tumours.
- **Paravertebral / Posterior** — spine, cord, neurovascular bundle, sympathetic chain. Masses: **neurogenic tumours (schwannoma, neurofibroma, meningocele)**, bony mets, paravertebral abscess, extramedullary haematopoiesis.

Symptoms

- **~50% asymptomatic** (incidental on imaging).
- Pressure effects: chest pain, cough, dyspnoea, wheeze, **fever/night sweats** (lymphoma), weight loss; specific syndromes (e.g. **thymoma + myasthenia gravis**).

Diagnosis

CT with 3D (mainstay), CXR, MRI, **EBUS/EUS**, mediastinoscopy, **FNA biopsy**, PET. Treatment: surgery (minimally invasive/robotic or sternotomy), chemo, radiation — by tissue type.

Special Signs

- **Mediastinal shift** — tracheal deviation (upper) / heart deviation (lower).
- **Mediastinal widening** — width **>8 cm** on CXR or mediastinum:chest ratio **>0.25** (think aortic injury/aneurysm).
- **Pneumomediastinum** — air tracking in (rupture, penetrating, iatrogenic, **Macklin effect**: alveolar rupture → air along bronchovascular sheath).

Mediastinitis

- Serious/life-threatening inflammation. **Subtypes: postoperative (commonest, after median sternotomy)**, descending necrotizing (neck/dental infection), fibrosing (chronic — TB, fungal, sarcoid).
- **Causes: oesophageal perforation** (Boerhaave, endoscopy, button battery), surgical, spreading infection.
- **Features:** severe pleuritic chest pain radiating to neck/back, fever, tachycardia, dyspnoea.
- **Diagnosis/treatment:** CT; **broad-spectrum antibiotics + surgical drainage + fix the cause** — urgent (sepsis risk).

Diaphragmatic Hernia (congenital)

- **Bochdalek** — commonest (75–90%), **posterior, left**.
- **Morgagni** — 2%, smaller, **anterior, right**.

Walk-Away Points

- **Anterior mass = 4 T's** (Thymoma, lymphoma, Thyroid, Teratoma); 50% of masses, thymoma ↔ myasthenia.
- **Middle = lymphadenopathy/lymphoma/cysts; posterior = neurogenic tumours.**
- **CT** is the workhorse; widening >8 cm = vascular emergency.
- **Mediastinitis** — postoperative or oesophageal perforation; antibiotics + drainage urgently.
- Bochdalek = **B**ack/left; Morgagni = anterior/right.

Mediastinum — Revision Table	
Boundaries	Detail
Anterior/posterior	Sternum / vertebrae
Superior/inferior	Thoracic inlet / diaphragm
Superior-inferior split	Sternal angle → T4-5 disc
Compartment (ITMIG)	Contents
Prevascular (anterior)	Thymus, fat, nodes
Visceral (middle)	Heart, aorta, SVC, trachea, oesophagus, thoracic duct
Paravertebral (posterior)	Spine, cord, sympathetic chain
Compartment	Typical mass
Anterior (~50%)	4 T's: Thymoma, lymphoma, Thyroid, Teratoma
Middle	Lymphoma, lymphadenopathy, cysts, aneurysm
Posterior	Neurogenic (schwannoma, neurofibroma)
Special sign	Definition
Mediastinal shift	Tracheal (upper) / heart (lower) deviation
Widening	>8 cm CXR or ratio >0.25
Pneumomediastinum	Air tracking; Macklin effect
Thymoma association	Myasthenia gravis
Mediastinitis	Detail
Commonest	Postoperative (sternotomy)
Key cause	Oesophageal perforation (Boerhaave)
Features	Chest pain → neck/back, fever
Treatment	Antibiotics + drainage + fix cause
Congenital diaphragmatic hernia	Site
Bochdalek	75–90%, posterior, left
Morgagni	2%, anterior, right
Diagnosis	CT 3D, EBUS/EUS, mediastinoscopy, FNA, PET

Bold = high-yield.